

Waraitip Laosangprateep

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EEC Chonburi-Rayong · Available from August 2026

SCAN ME



PROFILE

Chemical Engineering graduate with a Master's in Medical Engineering, bringing hands-on expertise in materials characterisation, process engineering, and experimental research across Taiwan and New Zealand. Proficient in spectroscopic techniques, DOE methodology, and GMP-aligned manufacturing environments. Bilingual (Thai/English) with strong APAC cultural fluency. Seeking a process or materials engineering role within a global chemicals or advanced materials company in the EEC zone, where a rigorous technical foundation supports manufacturing excellence and product innovation.

EDUCATION

Master of Medical Engineering · **University of Auckland** · New Zealand Mar 2025 – Jun 2026

- Thesis: Mechanical characterisation of growth plate structures using microindentation techniques
- Coursework: Advanced Functional Materials · Medical Device & Technology Development · Advanced Drug Disposition & Kinetics · Advanced Imaging Technologies · Engineering Project Management

Bachelor of Engineering – Chemical Engineering · **Thammasat University (SIIT)** · Thailand Aug 2020 – May 2024

- GPA: 3.22 / 4.00 · International Programme, full English instruction
- Senior Project: Theoretical study on butylone inclusion complexes with β -cyclodextrin — presented at PACCON 2024
- Core coursework: Chemical Reaction Engineering · Process Design · Thermodynamics · Heat & Mass Transfer · Fluid Mechanics

RESEARCH & INDUSTRY EXPERIENCE

Master's Thesis Researcher · **University of Auckland** · New Zealand Mar 2025 – Present

- Performed nanoindentation and microindentation mechanical testing on biological tissue samples across growth plate sub-structures
- Applied critical analysis to reconcile conflicting experimental data and form evidence-based conclusions
- Developed expertise in precision instrumentation, sample preparation, and quantitative data interpretation

Research Intern – SWCNTs Defect Characterisation · **Academia Sinica (IAMS-IIP)** · Taiwan Jun 2023– Jul 2023

- Conducted independent experiments on Single-Walled Carbon Nanotubes using absorbance and fluorescence spectroscopy for cancer diagnostic applications
- Designed and executed DOE-style experiments; analysed spectral data and contributed to peer scientific reports
- Gained proficiency in nanomaterial characterisation, precision instrumentation, lab safety protocols, and scientific communication

TECHNICAL SKILLS

Process Engineering: Chemical & process engineering fundamentals; thermodynamics; heat & mass transfer; reaction engineering; fluid mechanics; process scale-up principles

Materials & Analysis: Absorbance & fluorescence spectroscopy; nanoindentation; nanomaterial characterisation; optical microscopy principles

Quality & Standards: GMP manufacturing exposure; MSDS/SDS chemical safety documentation; quality control checkpoint documentation; factory compliance awareness

Process Improvement: Lean manufacturing awareness; Statistical Process Control (SPC) fundamentals; Design of Experiments (DOE); FMEA (Failure Mode & Effects Analysis); FTA (Fault Tree Analysis); Six Sigma concepts

Simulation & Tools: AspenPlus (process simulation); Gaussian / GaussView; AutoDock; Discovery Studio Visualizer; Origin Lab; WebLabViewer

Data & Computing: Python (data analysis, visualisation, supervised ML); statistical analysis; MS Office Suite (Word, Excel, PowerPoint, OneNote)

Regulatory: Biomedical waste management; lab biosafety (PHAC certified, 10+ certs); regulatory documentation awareness; technical report writing

ADDITIONAL EXPERIENCE

Customer Service & Operations · **Lily's Collection** · **Blue Elephant** · **Mi&Chi** · **Khao San Eatery** · **The Coffee Club** · Auckland, NZ Aug 2024 – Jan 2026

- Developed stakeholder communication, cross-cultural teamwork, and operational reliability skills transferable to customer-site engineering contexts

VOLUNTEER & LEADERSHIP

Narrator & Lab Demonstrator · **Thammasat University – SIIT Open Days** · Thailand Jul 2022 – May 2024

- Facilitated chemical process demonstrations for prospective students; contributed to increased enrolment through clear technical communication

CERTIFICATIONS & ACHIEVEMENTS

- Public Health Agency of Canada Certifications (10+): Biomedical Waste Management · Chemical Disinfectant · Pathogen Risk Assessment · Decontamination Technologies & Lab Biosafety
- Poster Presentation Certificate – PACCON 2024 (Pure and Applied Chemistry International Conference)
- Internship Certificate – Institute of Atomic and Molecular Sciences, Academia Sinica, Taiwan

LANGUAGES

Thai (Native) · English (IELTS 6.5 · TOEIC 830 — Professional) · Japanese (Elementary)